Parity, Utility and Safety Are Not Independent: A Controlled Comparison of Three Bias-Mitigation Families for Toxicity Classification

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Abstract*—Toxicity classifiers over-flag benign text that mentions demographic groups, and a large literature proposes fixes at three different points of the pipeline: the data, the loss, and the decision threshold. Those fixes are almost never compared against one another on one corpus under one recipe. We run that comparison. On the Jigsaw Unintended Bias corpus we fine-tune DistilBERT and evaluate eight trained configurations, plus a training-free post-processing arm, on three axes kept deliberately separate—utility, parity, and safety—each against a structurally controlled baseline that differs from it in exactly one thing. Identity-aware loss reweighting reduces the subgroup false-positive-rate gap by 0.0520 (95% CI $[-0.0742, -0.0297]$; $8/8$ seeds; exact sign test $p = 0.0078$) and flattens the regression of subgroup FPR on subgroup toxic rate by 44.6%, but costs 0.1138 of subgroup false-negative rate: 2.19 points of missed abuse per point of gap closed. Counterfactual augmentation is a null result at two doses, and the higher dose is the more informative one—it flattens the same shortcut by 30.9% while leaving the gap slightly worse, showing that removing the lexical shortcut is *not* sufficient to remove the disparity. Group DRO, the same objective with the human choice removed, moves the gap $2.6\times$ the wrong way and its learned weights explain why. Together with two hand-built controls this yields the paper's main claim: the parity gain requires an *asymmetric* weighting of the identity-bearing non-toxic cell alone, and any objective that also raises the identity-bearing toxic cell—chosen or learned—travels the same frontier backwards. Code, all 38 runs and every table: https://github.com/nigarrustamova/fairtox (tag `v1.0-final`).

***Index Terms*—algorithmic fairness, toxicity detection, content moderation, unintended bias, loss reweighting, counterfactual data augmentation, distributionally robust optimisation**

I. Introduction

A content-moderation classifier can fail in two directions, and the two failures land on different people. A *false negative* lets abuse through, and the person targeted by it is unprotected. A *false positive* removes a harmless comment, and the person who wrote it is silenced. When false positives concentrate on comments that merely *mention* a demographic group—"I am a proud Muslim woman"—the moderation system becomes a tax on the communities it was deployed to protect.

The mechanism is well understood in outline. Identity terms co-occur with abuse in scraped training text, because people name the groups they attack. Empirical risk minimisation is therefore free to treat the identity token itself as evidence of toxicity, which is a much cheaper rule to learn than the semantics of hostility [1], [2]. What is far less settled is what to do about it. Proposals exist at all three points where a machine-learning pipeline can be modified—rewrite the *data* [1], [4], change the *loss* [9], [10], or move the *decision threshold* [11]—and they are usually evaluated one at a time, against a baseline of the authors' own construction, on a metric of the authors' own choosing.

This paper runs the comparison those papers do not, around one question:

> *Given one corpus, one architecture and one training recipe, how do mitigations at the three intervention points compare on parity—and what does each of them cost in utility and in safety?*

We hold the corpus, the architecture, the recipe, the split, the seed and the metric set fixed, and vary only the mitigation. The design is deliberately conservative in one respect that matters: the baseline and every mitigated arm are produced by *the same function reading the same configuration file*, so a reviewer verifies the control by reading one function rather than by diffing two configurations.

Contributions:

1) A controlled comparison of **five mitigations across the three intervention points** plus **three controls**, on one corpus with one recipe: 38 training runs, 115 models, 34.4 GPU-hours (Section VI).
2) A replicated headline result. Loss reweighting reduces the subgroup FPR gap by 0.0520 over **eight seeds**, with a paired interval that excludes zero and an exact sign test at $p = 0.0078$, at a measured safety cost of 2.19 points of subgroup FNR per point of gap closed (Section VI-A).
3) A **measured mechanism**. Subgroup FPR is an almost linear function of the subgroup's toxic rate ($r = 0.885$–0.925 in every configuration we ran, the single most reproducible fact in the study). The intervention flattens that line by 44.6% without breaking it (Section VI-C).
4) A **dose-response null result** for counterfactual augmentation that is more informative than a failure. At full

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dose the augmentation demonstrably does its job—the shortcut flattens by 30.9%—and the disparity survives anyway. Removing the lexical shortcut is not sufficient to remove the disparity (Section VI-E).

5) The paper's central claim, established from three independent directions: **the parity gain requires an asymmetric weighting**. Ordinary class weighting, an isolated identity-toxic weighting, and Group DRO—which *learns* the weight vector and so contains no human choice at all—each move the gap in the wrong direction (Sections VI-F and VI-H).
6) An honest account of what the measured false positives are. A uniform random sample of 60 identity-bearing false positives was read and coded by hand: 45% are comments a reasonable annotator could have labelled toxic, and 38% are counter-speech, quotation or plain neutral mention (Section VI-K).

Everything is reproducible from a clean checkout by one command, and every number in every table below is generated from committed artefacts rather than typed: https://github.com/nigarrustamova/fairtox (tag `v1.0-final`).

II. Related Work

A. Measuring unintended bias

Dixon et al. [1] introduced the problem in its modern form, showing that toxicity classifiers assign elevated scores to benign sentences containing identity terms, and proposing a synthetic template test set to measure it. Borkan et al. [2] argued that a single aggregate hides the phenomenon and proposed threshold-agnostic per-subgroup AUCs—Subgroup AUC, BPSN (Background Positive, Subgroup Negative) and BNSP—together with the Jigsaw Unintended Bias corpus with crowd-sourced identity annotations that we use here. Sap et al. [3] showed the same failure along a dialect axis: annotators' insensitivity to African-American English produces corpora on which models flag AAE tweets up to twice as often. Park et al. [4] documented the gender version. Garg et al. [5] survey the field and observe that most work reports a single mitigation against a single baseline; Gupta et al. [6] study the accuracy–fairness frontier explicitly and find that no single point on it dominates. Our three-axis reporting follows that last observation directly.

B. Pre-processing: change the data

The oldest family swaps identity terms in the training text so that the token stops predicting the label. Dixon et al. [1] balanced their corpus by adding benign examples for over-represented terms; Park et al. [4] used gender swapping; Garg et al. [7] formalised the idea as counterfactual token fairness, requiring a model's score to be invariant under substituting one identity term for another. The load-bearing assumption is that toxicity *is* invariant under such substitution, which is close to true for abuse and plainly false for statements of fact. Recent work continues in this direction with causal framing [8]. We implement the classic form and report it at two doses.

C. In-processing: change the objective

The second family modifies training. Zhang et al. [9] used an adversary that predicts the protected attribute from the model's output, and penalised the main model for making that easy. Sagawa et al. [10] proposed Group DRO, which minimises the worst-group loss by learning an online weight vector over pre-declared groups. Per-sample loss weighting—upweighting the cell that breaks the term–label correlation—is the simplest member of the family and the one we take as our headline method. We include Group DRO because it is the same intervention with the human choice removed, which makes it a control rather than merely a neighbour.

D. Post-processing: change the decision

The third family leaves the trained model untouched and moves the decision threshold per group. Hardt et al. [11] give the classical construction for equalised odds and show it is optimal among post-hoc adjustments. It is cheap—it needs no training at all—and it carries a deployment cost the other two families do not: the group membership of an input must be known *at inference time*. We measure it and report that cost wherever we quote it.

E. What this paper adds

To our knowledge no prior study evaluates all three intervention points on one corpus, with one architecture, one recipe, one metric set, and replication across seeds, while separating utility, parity and safety rather than collapsing them. The comparison is what produces our central claim: the three families do not merely differ in strength, they differ in *sign*, and the reason is a structural property of the weight vector rather than a property of any one method.

III. Problem Formulation

A. Two errors, three axes

Let $y \in \{0,1\}$ be a binarised toxicity label and $\hat{p} \in [0,1]$ a model's score, thresholded at τ. For a subgroup k—the set of comments annotated as mentioning group k—we report

$$\mathrm{FPR}_k = \Pr(\hat{p} \geq \tau \mid y=0,\ k), \tag{1}$$

$$\mathrm{FNR}_k = \Pr(\hat{p} < \tau \mid y=1,\ k). \tag{2}$$

FPR_k is the wrongful-censorship rate for group k and is the study's central quantity; FNR_k is the rate at which genuine abuse aimed at group k is missed.

Results are reported on three axes that are never combined into one score, because a single "is it better?" number would hide exactly the trade-off the study exists to measure:

- **utility**: macro-F1, ROC-AUC, PR-AUC;
- **parity**: the FPR *gap* $\max_k \mathrm{FPR}_k - \min_k \mathrm{FPR}_k$, the macro FPR $\frac{1}{K}\sum_k \mathrm{FPR}_k$, the FPR variance, and the equalised-odds difference $\max_k |\mathrm{FPR}_k - \mathrm{FPR}|$;
- **safety**: macro FNR over the subgroups entitled to one, and the FNR gap.

We additionally report the threshold-free BPSN and BNSP AUCs of Borkan et al. [2], so that no conclusion rests on the choice of τ.

B. Three guards on the numbers

Three properties of this setting will silently corrupt a fairness result, so each is handled explicitly.

Separate support floors. FPR_k is estimated on group k's non-toxic rows and FNR_k on its toxic rows, and in this corpus those counts differ by an order of magnitude. A single floor would either admit an FNR computed from eleven comments or discard a well-supported FPR. Each rate therefore carries its own floor ($N \geq 100$) and its own reportable flag, and the two aggregates are computed over *different* subgroup sets: 14 subgroups can carry a parity claim and only 9 a safety claim.

Wilson intervals. At zero observed false positives, a percentile bootstrap returns $[0, 0]$ whether the subgroup has 120 non-toxic rows or 12,000, because resampling a vector of zeros always yields zero. The Wilson score interval [18] still widens as the sample shrinks, which is the correct behaviour in precisely the case a fairness paper is most likely to overstate. Every per-subgroup rate we report carries one.

Collapse detection. A model that flags nothing has $\mathrm{FPR}_k = 0$ in every subgroup and therefore *perfect* parity. Read without a check, a dead model is the fairest model in any study of this kind. Every arm is tested for single-class output; a collapsed arm is flagged, excluded from the verdict, and marked on every figure it appears in. No arm reported in this paper collapsed.

C. Pre-registered outcomes

The interpretation of each possible result was fixed in a configuration file before any model was trained: **A**, gap falls at a macro-F1 cost within tolerance; **B**, gap falls and aggregate performance falls with it; **C**, gap falls but subgroup FNR rises beyond tolerance—a safety regression; **D**, null or adverse. All four are reportable findings. Only a collapsed model is a failure, and it is a failure to be fixed rather than written up.

IV. Method

A. The intervention

Training examples are partitioned into four cells by (identity-bearing) × (label). The headline method attaches a per-sample weight to the binary cross-entropy loss, raising *only* the cell that breaks the term–label correlation: identity-bearing and non-toxic. Writing w_i for the weight of example i and ℓ_i for its unweighted loss,

$$\mathcal{L} = \frac{\sum_i w_i \, \ell_i}{\sum_i w_i}, \qquad w_i = \begin{cases} \alpha & \text{identity-bearing, } y_i = 0, \\ 1 & \text{otherwise,} \end{cases} \tag{3}$$

with $\alpha = 4$ fixed in advance and swept in the ablation.

The normalisation in (3) is by the *sum of weights*, not the batch size. Dividing by the batch size would make the gradient magnitude grow with α, so α would act partly as a learning rate and the ablation would measure that confound rather than the weighting scheme. This one-line choice is what makes the sweep interpretable.

Identity annotations are used in exactly two places: to compute training weights, and to slice the evaluation. The model never receives a demographic feature, at training time or at inference.

Algorithm 1 The controlled pair

Require: corpus D, config C, seed s, mitigation m

$D_{\text{tr}}, D_{\text{va}}, D_{\text{te}} \leftarrow \textsc{Split}(D, s)$ ▷ stratified on $y\times$ identity
$\phi \leftarrow \textsc{Fingerprint}(D_{\text{tr}}, D_{\text{va}}, D_{\text{te}})$
for all $a \in \{\textsc{baseline}, \textsc{mitigated}\}$ **do**
  $X_a \leftarrow D_{\text{tr}}; \quad w_a \leftarrow \mathbf{1}$
  if $a = \textsc{mitigated}$ **then**
    apply m: reweight w_a, **or** rewrite X_a, **or** replace the objective
  end if
  $\theta_a \leftarrow \textsc{Train}(X_a, w_a, C, s)$ ▷ one function, both arms
  $P_a \leftarrow \textsc{Predict}(\theta_a, D_{\text{va}} \cup D_{\text{te}})$
end for
assert $\phi_{\text{BASE}} = \phi_{\text{MIT}}$ and $\textsc{TrainFp}_{\text{BASE}} = \textsc{TrainFp}_{\text{MIT}}$
assert neither P_a is single-class
return per-axis differences on D_{te}

B. The controlled comparison

Algorithm 1 states the protocol. Its important property is structural rather than procedural: $\textsc{Train}$ is one function and the two arms call it with one differing argument, so there is nowhere for a second difference to enter. Three guards then run before any comparison is written. Both arms must share a *split fingerprint*, a hash of the identifier-to-split assignment. Both must share a *training fingerprint*, a hash of every setting that is not the variable under test; a comparison whose arms disagree is refused rather than reported. Differing GPUs produce a warning, because a headline pair split across two machines carries a hardware confound inside the very comparison the study rests on.

C. The five mitigations and three controls

Each configuration below changes *one* thing relative to the headline setup; the recipe is never retuned, because retuning would confound a method comparison with a tuning difference.

- **Reweighting** (in-processing, headline): (3) with $\alpha = 4$.
- **Counterfactual swap** (pre-processing): identity terms in the training text are replaced by terms for another group on the same axis, consistently within a comment, at probability $p \in \{0.5, 1.0\}$. Swaps are made *in place*, never appended: appending would change the step count, the wall-clock cost and the effective epoch count at once.
- **Group DRO** (in-processing): the same four cells, but the weight vector is learned online [10] instead of chosen.
- **Per-subgroup thresholds** (post-processing): a threshold per subgroup, fitted on validation and scored on test, following the false-positive side of [11].
- **Held-out axes**: the weighting computes its identity signal from gender and religion only, while the audit continues to measure every subgroup. The 14 reportable subgroups split 7/7 by axis, decided before the run.

TABLE I
CORPUS COMPOSITION AND REPORTING COVERAGE

Quantity	Rows	Toxic rate	Identity rate
Comments in the release	1,999,516	–	–
Identity-annotated subset	447,998	0.1134	0.4189
train	335,998	–	–
val	44,800	–	–
test	67,200	–	–
Subgroups configured	24	–	–
clearing the FPR floor	14	–	–
clearing the FNR floor	9	–	–

Three controls complete the design. The **`class arm`** weights by inverse class frequency and knows nothing about identity: it answers "would ordinary imbalance handling have done this?" The **isolated toxic cell** raises the identity-bearing *toxic* weight alone. The **fixed-epoch arm** keeps the final epoch in both arms instead of the best, removing checkpoint selection from the comparison.

V. EXPERIMENTAL SETUP

A. Corpus and splits

We use the Jigsaw Unintended Bias in Toxicity Classification corpus [14], released under CC0 1.0. Table I gives the composition. Two decisions shape every number downstream. First, the released target is the *fraction* of annotators who called a comment toxic; we binarise at 0.5, and because that is an assumption rather than a fact we re-run the entire audit at $\tau \in \{0.3, \dots, 0.7\}$. Second, we use only the identity-annotated subset: a row with no identity rating is *unrated*, not identity-free, and conflating the two would sweep genuinely identity-bearing comments into the background weight and corrupt the intervention.

The split is 75/10/15, stratified on label $\times$ any-identity, made once before any statistic is fitted, and fingerprinted. The unusually large test fraction is deliberate: the reporting floor is counted on the test split, so 15% instead of 10% costs about 6% of training throughput and buys roughly 50% more subgroups that the paper can make a claim about.

B. Model and training

The classifier is a pretrained encoder with a single-logit head: `[CLS]` $\rightarrow$ dropout $\rightarrow$ `Linear`$(768, 1)$. We use DistilBERT [12] for the headline study and DistilRoBERTa—a RoBERTa [13] distilled by the procedure of [12]—for the backbone-generalisation arm. The two differ in pretraining corpus and in tokeniser (byte-level BPE rather than WordPiece), which is what makes the comparison informative. The head reads position 0 of the last hidden state, which is defined identically for both, so swapping the backbone required no code change. We deliberately do not use `AutoModelForSequenceClassification`: it hides the loss, and every intervention in this study lives in the loss.

Training uses AdamW, learning rate 3×10^{-5}, batch size 64, three epochs, warmup ratio 0.1, weight decay 0.01, gradient clipping at 1.0, maximum sequence length 128, and mixed precision. Model selection is on validation macro-F1 with early stopping (patience 2)—applied identically to both arms.

C. Compute

All 115 models were trained on one MIG 3g.20gb slice of an NVIDIA A100-SXM4-40GB (19.5 GB visible), for 34.4 GPU-hours in total. Peak reserved memory was 3.99 GB, a fifth of the allocation; at batch 64 the model is already compute-bound, so the spare memory was spent on additional seeds rather than on a larger batch, which would have halved the optimiser steps.

D. Implementation and reproducibility

Everything is implemented in Python with PyTorch [16] and the Hugging Face Transformers library [15] for the encoders, and scikit-learn [17] for the classical baseline, the splitting and the aggregate metrics. The whole study is one command from a clean checkout, and the environment is pinned by a lock file generated on the machine that produced the reported numbers.

Seeds are fixed for Python, NumPy and Torch, and the driver reseeds per stage so a stage's result does not depend on which stages ran before it. Every run records an environment manifest, a configuration hash and the two fingerprints above. Continuous integration runs lint, the full test suite (379 tests) and the complete pipeline on synthetic data on every push, and re-renders every figure from committed metrics. Every table in this paper is emitted from those committed metrics by a script rather than typed, so the paper cannot drift away from the results it reports.

We also measured the noise floor. Five of our configurations share an *identical* baseline arm—the unweighted arm ignores every mitigation setting—so the same model was trained five separate times at seed 42. Those five runs land in exactly *two* states, not five, and the two are separated by a code revision rather than by the GPU: 46 of 67,200 test decisions differ (0.068%). The resulting measurement floor is 0.0000 on the FPR gap and 0.0017 on the worst-moving metric, roughly $78\times$ smaller than the effect we report. The seed-to-seed spread in Section VI is therefore genuine seed variance, not run-to-run noise.

VI. RESULTS

A. The headline pair

Table II reports the controlled pair over eight seeds. Every parity metric falls, macro-F1 falls slightly, and subgroup FNR rises substantially. All eight seeds agree in sign on all five metrics, all eight land on the same pre-registered outcome—**C: safety trade-off**—and no model collapsed.

The exchange rate is the number to quote when someone asks what the intervention costs: $0.1138/0.0520 = 2.19$ points of subgroup false-negative rate per point of FPR gap closed. Neither arm is a good moderation system in absolute terms—the baseline already misses 42% of genuine abuse—so our

TABLE II

THE CONTROLLED PAIR, EIGHT SEEDS. COLUMNS ARE MEANS OVER SEEDS; Δ IS THE MEAN OF THE *per-seed* DIFFERENCES, WHICH IS WHAT MAKES THE INTERVAL AND THE SIGN TEST PAIRED. THE SIGN COLUMN COUNTS SEEDS MOVING IN THE DIRECTION THE ROW IS ABOUT.

Axis	Metric	Baseline	Mitigated	Δ	95% CI	Sign	p
Parity	FPR gap	0.1072	0.0552	-0.0520	$[-0.0742, -0.0297]$	8/8	0.0078
	macro FPR	0.0695	0.0372	-0.0323	$[-0.0451, -0.0196]$	8/8	0.0078
	EOD (FPR)	0.0962	0.0413	-0.0550	$[-0.0790, -0.0310]$	8/8	0.0078
Utility	macro-F1	0.8031	0.7903	-0.0128	$[-0.0153, -0.0104]$	8/8	0.0078
Safety	subgroup FNR	0.4382	0.5520	$+0.1138$	$[+0.0824, +0.1452]$	8/8	0.0078

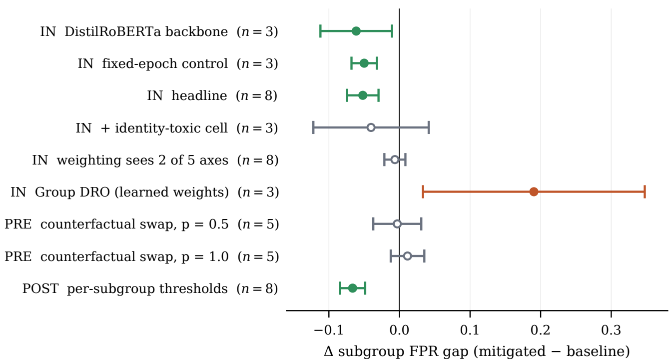

Fig. 1. Change in the subgroup FPR gap for every configuration, with the paired 95% confidence interval. A filled marker means the interval excludes zero. Green is a significant improvement, orange a significant deterioration, grey inconclusive. IN, PRE and POST mark the intervention point. Three-seed families carry a direction and an interval but no significance claim; their sign test cannot fall below 0.25.

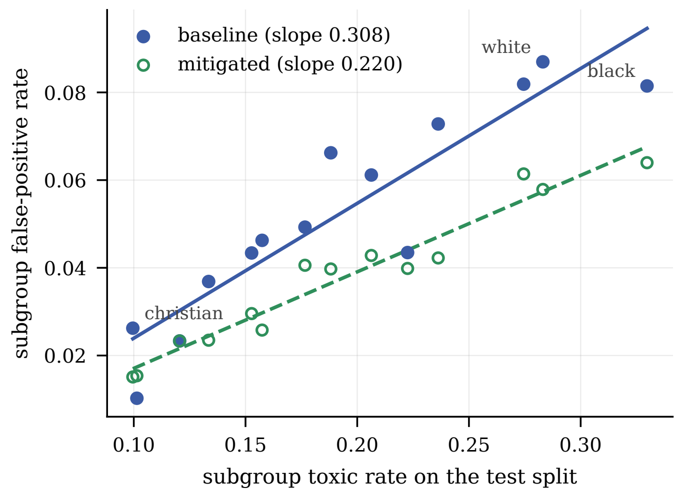

Fig. 2. Subgroup false-positive rate against the subgroup's toxic rate on the test split, seed 42, over the 14 reportable subgroups. Lines are least-squares fits drawn only across the range their own data covers. Over the eight seeds the slopes are $0.4405 \to 0.2310$, a 44.6% flattening, with $r = 0.9043 \to 0.9098$.

claim is about the *controlled difference*, not about absolute moderation quality.

For reference, a TF-IDF and logistic-regression baseline, deliberately strengthened with balanced class weights, reaches macro-F1 0.7460 and ROC-AUC 0.9096 against DistilBERT's 0.8003 and 0.9387. The two sit at very different operating points, however—the classical model misses half as much abuse (FNR $= 0.2352$ vs. 0.4211) while flagging four times as many benign comments (FPR $= 0.1179$ vs. 0.0289)—so we quote both rates and never macro-F1 alone.

B. All eight configurations

Fig. 1 and Table III place every configuration of Section IV-C on one axis. The three intervention points do not merely differ in strength: they differ in sign.

Reweighting works, and works on both backbones—on DistilRoBERTa the baseline gap is *larger* (0.0866 vs. 0.0767 at seed 42) although the model is better on every quality metric, and the mitigation is more effective there, so the phenomenon is not an artefact of one pretrained model.

The fixed-epoch control reproduces the effect at -0.0500 and collapses its spread. Compared like for like on the same three seeds, the standard deviation of the per-seed difference falls from 0.0427 to 0.0072, a factor of 5.9. The two arms do not peak on the same epoch—across those seeds the baseline was kept at epoch $1, 1, 2$ and the mitigated arm at $2, 2, 2$—so the kept models differed in more than the weight vector. Removing that difference leaves the effect intact and locates most of the variance in checkpoint selection rather than in the mitigation.

C. The mechanism

Fig. 2 shows why the disparity exists. Across the reportable subgroups, a group's toxic rate on the test split predicts the model's FPR on that group almost perfectly. Over the eight headline seeds the baseline slope is 0.4405 with $r = 0.9043$; the mitigated slope is 0.2310 with $r = 0.9098$. The intervention **flattens the line by** 44.6% **and does not break it**: the residual fit is just as tight. The model is not "prejudiced" against groups; it has learned a conditional probability from the training distribution and applies it as a prior to individual benign comments.

Table IV fits the same regression in every configuration. Two facts follow. The baseline slope is 0.44–0.48 and the baseline correlation 0.885–0.925 *everywhere*, which makes the shortcut the most reproducible observation in the study. And the slope change does not track the gap change: Fig. 3 shows

TABLE III

EVERY CONFIGURATION ON THE THREE AXES. ONLY THE EIGHT-SEED ROWS CARRY A SIGNIFICANCE CLAIM. A SAFETY TAX IS PRINTED ONLY WHERE THE GAP CHANGE ITSELF EXCLUDES ZERO; ELSEWHERE THE RATIO WOULD DESCRIBE A NEAR-ZERO DENOMINATOR.

Stage	Configuration	n	Δ gap	sd	95% CI	Sign	Δ F1	Δ FNR	Tax
IN	loss reweighting (headline)	8	-0.0520	0.0266	$[-0.0742, -0.0297]$	8/8	-0.0128	$+0.1138$	2.19
IN	reweighting, fixed epoch	3	-0.0500	0.0072	$[-0.0680, -0.0321]$	3/3	-0.0111	$+0.1306$	2.61
IN	reweighting, DistilRoBERTa	3	-0.0613	0.0204	$[-0.1120, -0.0107]$	3/3	-0.0168	$+0.1526$	2.49
IN	reweighting + identity-toxic cell	3	-0.0403	0.0329	$[-0.1220, +0.0414]$	3/3	-0.0048	$+0.0698$	–
IN	reweighting, 2 of 5 identity axes	8	-0.0065	0.0178	$[-0.0214, +0.0084]$	4/8	-0.0088	$+0.0830$	–
IN	Group DRO, learned weights	3	$+0.1904$	0.0632	$[+0.0332, +0.3475]$	0/3	-0.0472	-0.2308	–
PRE	counterfactual swap, $p = 0.5$	5	-0.0031	0.0274	$[-0.0371, +0.0309]$	4/5	-0.0014	-0.0016	–
PRE	counterfactual swap, $p = 1.0$	5	$+0.0114$	0.0192	$[-0.0124, +0.0352]$	1/5	-0.0069	$+0.0450$	–
POST	per-subgroup thresholds	8	-0.0664	0.0213	$[-0.0842, -0.0486]$	8/8	-0.0128	$+0.1183$	1.78

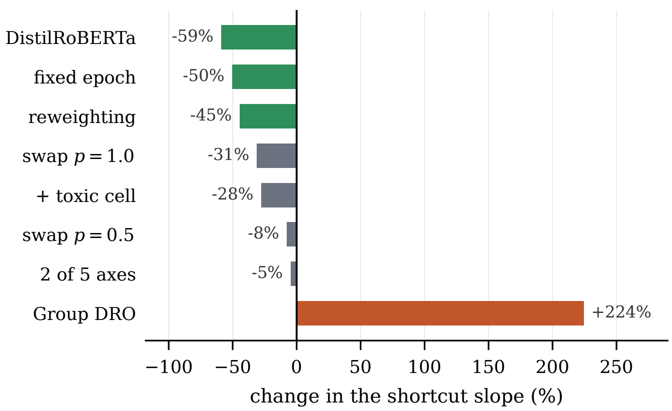

Fig. 3. How far each configuration moves the shortcut slope. Green marks the configurations that also close the FPR gap. The two grey bars at -31% and -28% move the mechanism substantially *without* closing the gap, and Group DRO makes the shortcut three times steeper.

two configurations that move the shortcut a long way without closing the gap at all.

D. Is the gain only a threshold shift?

The first objection to any parity result of this kind is that the mitigated model simply flags less. We test it directly: for each of the mitigated model's operating points we interpolate the baseline's gap *at the same global false-positive rate*, and compare there. Points outside the baseline's measured range are reported as non-comparable and excluded rather than clamped.

Over the eight seeds this yields 36 comparable points. The mitigated model has a smaller gap at 33 of them (91.7%), with a mean reduction of 23.9% and a median of 26.0%. Three points go the other way, all at high thresholds in seeds 47 and 49 (-14.6%, -3.2%, -0.2%). The effect therefore survives matching at almost every operating point but not at all of them, which closes the objection while reporting the exceptions.

Relatedly, the mitigated model's parity depends far less on where the operator sets the dial: across $\tau \in [0.3, 0.7]$ the baseline's gap spans 0.1621 and the mitigated model's 0.0425, $3.8\times$ narrower. For a deployed system whose threshold is retuned continually, that stability is a practical result in its own right.

Finally, direction. Because the gap is $\max - \min$, it falls both when the worst-off subgroup improves (*levelling up*) and when the best-off one deteriorates (*levelling down*), and only the first is a fairness gain. Over the eight headline seeds the closure is levelling up in six and mixed in two. By contrast Group DRO levels down in $3/3$ and counterfactual swapping in $4/5$.

E. Counterfactual augmentation: a dose-response null

At three seeds, counterfactual swapping looked like a small win. At five it is -0.0031 with an interval spanning zero (Table III); the earlier verdicts were a small-sample artefact. Doubling the dose settles it. At $p = 1.0$, every identity mention in training is randomised, and the gap moves the *wrong* way ($+0.0114$, improving in $1/5$ seeds). **A method whose effect does not grow with its dose is not doing what its mechanism claims.**

The reason is sharper than "it does nothing", and it is the most useful negative result in the paper. At $p = 1.0$ the augmentation *demonstrably works*: the shortcut regression flattens by 30.9% (Fig. 3), so the model measurably stops using the identity token to predict toxicity. The disparity survives anyway. **Removing the lexical shortcut is not sufficient to remove the disparity**—the token was never the whole cause. The threshold-free metrics agree: BPSN moves by -0.0026 at $p = 0.5$ and -0.0012 at $p = 1.0$, which is nothing at either dose.

This also explains why reweighting succeeds where swapping does not. Reweighting flattens the same slope by 44.6% *and* closes the gap, because it prices the whole identity-bearing region of input space rather than editing one token inside it.

F. Group DRO: the objective refutes itself

Group DRO learns a weight vector over the same four cells we weight by hand, which makes it our method with the one human decision removed. It reached outcome **D** in $3/3$ seeds: the FPR gap became $2.6\times$ worse ($0.1153 \to 0.3056$) while subgroup FNR fell by 0.2308. This is not a collapse but a real, well-behaved model that travelled the frontier backwards.

TABLE IV
THE SHORTCUT REGRESSION $\mathrm{FPR}_k = \beta\, t_k + c$, WHERE t_k IS SUBGROUP k'S TOXIC RATE, FITTED OVER THE REPORTABLE SUBGROUPS IN EVERY CONFIGURATION. "GAP CLOSES?" IS TAKEN FROM TABLE III: YES ONLY WHERE THE PAIRED INTERVAL EXCLUDES ZERO IN THE FAIR DIRECTION.

Configuration	n	Baseline slope	r	Mitigated slope	Change	Gap closes?
reweighting, DistilRoBERTa	3	0.467	0.916	0.189	−59.0%	yes
reweighting, fixed epoch	3	0.442	0.887	0.219	−50.4%	yes
loss reweighting (headline)	8	0.441	0.904	0.231	−44.6%	yes
counterfactual swap, $p = 1.0$	5	0.466	0.885	0.319	−30.9%	no
reweighting + identity-toxic cell	3	0.481	0.925	0.323	−27.7%	no
counterfactual swap, $p = 0.5$	5	0.466	0.885	0.424	−7.6%	no
reweighting, 2 of 5 identity axes	8	0.440	0.904	0.403	−4.7%	no
Group DRO, learned weights	3	0.480	0.924	1.447	+224.4%	no

TABLE V
WHAT GROUP DRO LEARNED, AGAINST WHAT WE ASSUMED. THE LEARNED COLUMN GIVES THE FINAL WEIGHT VECTOR FOR SEEDS 42/43/44; OUR STATIC VECTOR IS SHOWN NORMALISED TO SUM TO ONE FOR COMPARABILITY.

Cell	Ours	Group DRO, per seed
background, non-toxic	0.143	0.000 / 0.000 / 0.000
background, toxic	0.143	0.142 / 0.146 / 0.163
identity, non-toxic	0.571	0.451 / 0.424 / 0.446
identity, toxic	0.143	0.407 / 0.431 / 0.391

Table V shows why. The optimiser zeroed the easy majority cell and split its weight almost evenly between the *two* identity cells—which is essentially the configuration we tested by hand and rejected, rediscovered without a human choosing it, and it reproduced that configuration's failure mode amplified.

A worst-group objective *cannot* find our solution, and the reason is structural: the identity-bearing toxic cell also carries high loss, so any worst-group criterion necessarily raises it.

G. Does the effect reach groups it was never paid to see?

If the intervention teaches "identity-bearing language deserves more care" it should transfer; if it teaches "these particular tokens are expensive" it should not. We test this by computing the weighting's identity signal from gender and religion only, while the audit continues to measure every subgroup (Table VI).

The intervention transfers at about 58% strength. The confound this could have had is measured and absent: the weighted axes improve by -0.0277 here against -0.0258 for the same axes in the full run, so narrowing the weighting did not weaken it where it still applied.

Two qualifications must travel with this result. First, the paired interval excludes zero only just, and a sign test on $6/8$ does not ($p = 0.29$); we call it suggestive, not established. Second, **the transfer is a level shift, not a slope change**: this configuration lowers macro FPR in $8/8$ seeds but flattens the shortcut by only 4.7%, against 44.6% for the full weighting. Teaching the loss about two axes makes the model more cautious about identity-bearing language in general without re-ordering the groups relative to their toxic rate. That is also why its FPR *gap* appears flat while almost every subgroup

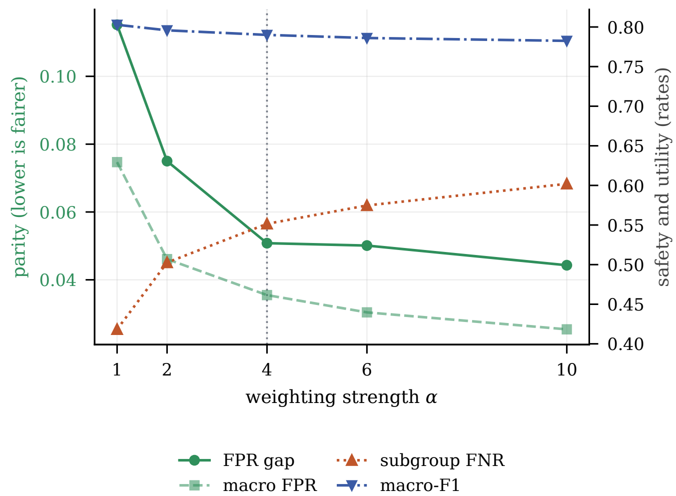

Fig. 4. The three axes as the weighting strength rises, three seeds. Parity gains flatten after the dotted line—the pre-registered $\alpha = 4$—while the safety cost keeps climbing. Macro FPR is shown alongside the FPR gap because the gap depends on only two subgroups and is the noisier of the two.

improves: the gap is $\max - \min$, and the max is a held-out group.

H. Ablation and the two mirror controls

Table VII sweeps α and reports the `class` control. Parity gains flatten after $\alpha = 4$ while the safety cost keeps climbing (Fig. 4), so the pre-registered default sits at the knee rather than at a tuned optimum. No arm collapsed at any α, including 10.

The `class` control is decisive. Weighting by inverse class frequency, which knows nothing about identity, makes the FPR gap $2.6\times$ *worse* ($0.1152 \to 0.3005$) while flagging 17.7% of everything. Ordinary imbalance handling is not a substitute for identity-aware weighting; it is the opposite of one.

The second control is its mirror. Isolating the identity-bearing *toxic* cell moves the FPR gap from 0.0767 to 0.1374 (worse) while macro FNR falls from 0.4744 to 0.3374 (better)—the exact reflection of the `class` result. Together the two controls turn "we observed a trade-off" into "the two axes are structurally coupled in this weighting family, and we demonstrated it from both sides".

TABLE VI

THE HELD-OUT EXPERIMENT OVER EIGHT SEEDS. `MAIN` IS SHOWN UNDER THE SAME AXIS SPLIT, WHICH REMOVES THE DOSE CONFOUND: THE WEIGHTED AXES IMPROVE BY ESSENTIALLY THE SAME AMOUNT IN BOTH RUNS, SO THE MOVEMENT ON THE UNTOUCHED AXES IS TRANSFER RATHER THAN A DILUTED DOSE.

Axis bucket	n	mean Δ FPR	sd	95% CI	Seeds	p	Subgroup obs.
`heldout`: weighted axes (gender, religion)	8	-0.0277	0.0162	$[-0.0412, -0.0142]$	8/8	0.0078	56/58
`heldout`: held-out axes (race, orient., disab.)	8	-0.0161	0.0185	$[-0.0315, -0.0007]$	6/8	0.2891	36/56
`main`: same weighted axes	8	-0.0258	0.0124	$[-0.0362, -0.0155]$	8/8	0.0078	56/58
`main`: same held-out axes	8	-0.0392	0.0193	$[-0.0553, -0.0230]$	8/8	0.0078	55/56

TABLE VII

ABLATION ON THE HEADLINE CONFIGURATION, THREE SEEDS. THE `CLASS` ARM IS THE CONTROL THAT ANSWERS WHETHER ORDINARY IMBALANCE HANDLING WOULD HAVE PRODUCED THE SAME EFFECT. THE FLAG RATE IS PRINTED BESIDE EVERY ARM BECAUSE AN ARM CAN POST A SMALL GAP SIMPLY BY FLAGGING ALMOST NOTHING.

Scheme	Arm	FPR gap	macro FPR	macro-F1	subgroup FNR	Flag rate
subgroup	$\alpha = 1$ (= baseline)	0.1152	0.0747	0.8028	0.4183	0.1050
subgroup	$\alpha = 2$	0.0750	0.0461	0.7958	0.5026	0.0917
subgroup	$\alpha = 4$	0.0508	0.0355	0.7900	0.5515	0.0846
subgroup	$\alpha = 6$	0.0501	0.0304	0.7861	0.5747	0.0817
subgroup	$\alpha = 10$	0.0443	0.0254	0.7825	0.6020	0.0787
class	inverse class frequency	0.3005	0.2157	0.7776	0.2039	0.1770

I. Threshold-free confirmation

Table VIII reports the Borkan metrics, which require no threshold at all. For the headline configuration BPSN—the threshold-free analogue of our FPR gap—rises from 0.8751 to 0.8865, while BNSP, which measures under-protection, falls from 0.9413 to 0.9244. Two independent metric families therefore agree with the thresholded result: the parity gain is real and so is the safety cost, and neither conclusion depends on $\tau = 0.5$.

One caveat deserves its own sentence, because it separates two claims that are easily conflated. **Group DRO improves BPSN** ($+0.0098$) while its thresholded FPR gap gets $2.6\times$ worse. Threshold-free parity and parity at a fixed operating point are different claims, and Group DRO is the case that separates them: it shifts the operating point so far that a better ranking still produces a much worse gap at $\tau = 0.5$.

J. Post-processing, and the cost it hides

Per-subgroup thresholds—fitted on validation, scored on test, with 13 or 14 of the 14 reportable subgroups receiving one—reduce the gap by 0.0664 in $8/8$ seeds (Table III). That is a *larger* parity gain than retraining, at no GPU cost at all.

It must never be quoted without the sentence that follows. A per-group threshold needs the identity annotation of a comment **at inference time**. The training-time methods consume identity annotations once, during training, and never again. In a deployed moderation system the post-processing method would require inferring or storing the demographic attributes of the text being judged—which is both a practical obstacle and an ethical one. The method wins on parity and loses on deployability, and both halves are the result.

K. What the false positives actually are

Aggregate rates say how much; they do not say what. We drew a *uniform random* sample of 60 of the baseline's $1{,}025$ identity-bearing false positives—deliberately not the most confident ones, whose composition is the composition of the confident tail rather than of the errors—and read and coded every one along two axes: whom the hostility is aimed at, and whether the dataset's non-toxic label is defensible (Table IX).

Two numbers change how the rest of the paper should be read.

27 **of** 60 (45%) **are comments a reasonable annotator could have called toxic.** Nearly half of what the metrics call a false positive is annotator disagreement rather than model error. Stating this does not weaken the study's claim—the claim is about a controlled difference between two arms measured identically—but it does bound what the absolute FPR means.

In the other half the model penalises exactly the text that should be protected. Counter-speech, quoted hostility and plain neutral mention together are $23/60$ (38%): text that argues against bigotry, quotes it in order to criticise it, or merely names a group. For a moderation system this is the worst error type available, because it silences the speech the policy exists to protect.

We also measured our own regular-expression triage against the human coding and report that it is weak: it recovers 3 of the 12 true counter-speech comments (25% recall) at 75% precision. Every category count in this paper therefore comes from the coded sample, never from the heuristic columns.

TABLE VIII
THRESHOLD-FREE METRICS OF BORKAN ET AL. [2], AVERAGED OVER EACH CONFIGURATION'S SEEDS. BPSN IS THE THRESHOLD-FREE ANALOGUE OF THE FPR GAP; BNSP MEASURES UNDER-PROTECTION.

Configuration	n	BPSN base	BPSN mit.	Δ BPSN	Δ BNSP	Δ AUC
loss reweighting (headline)	8	0.8751	0.8865	+0.0114	−0.0169	−0.0061
reweighting, DistilRoBERTa	3	0.8788	0.8941	+0.0153	−0.0170	−0.0046
counterfactual swap, $p = 0.5$	5	0.8738	0.8712	−0.0026	−0.0003	−0.0015
counterfactual swap, $p = 1.0$	5	0.8738	0.8726	−0.0012	−0.0072	−0.0042
Group DRO, learned weights	3	0.8746	0.8844	+0.0098	−0.0166	−0.0070
reweighting, 2 of 5 identity axes	8	0.8751	0.8770	+0.0019	−0.0157	−0.0057

TABLE IX
MANUAL CODING OF A UNIFORM RANDOM SAMPLE OF 60 IDENTITY-BEARING FALSE POSITIVES FROM THE BASELINE MODEL. "LABEL ARGUABLE" COUNTS COMMENTS A REASONABLE ANNOTATOR COULD HAVE CALLED TOXIC.

Category	n	%	Label arguable	What the comment is
`hostility_not_group_directed`	21	35.0	11	hostile, but aimed at a person, party or ideology
`group_directed_derogation`	16	26.7	16	genuinely derogatory towards the group
`counter_speech`	12	20.0	0	argues *against* bigotry
`identity_topic_neutral`	7	11.7	0	neutral mention, no hostility at all
`quoted_hostility`	4	6.7	0	quotes hostile speech in order to criticise it

L. Who is affected

Fig. 5 and Fig. 6 give the per-subgroup picture that the aggregates compress. Twelve of the 14 reportable subgroups improve on FPR; `asian` is unchanged—its false-positive rate is identical in both arms to six decimal places—and `atheist` gets *worse* (0.0103 → 0.0154, +50%). The most strongly supported single claim in the study is that a benign comment mentioning `white` is flagged 3.3× more often than one mentioning `christian` (0.0870 vs. 0.0262; thousands of rows each, disjoint Wilson intervals). On the safety side, all nine subgroups entitled to an FNR claim get worse, and strikingly evenly—between 7.5 and 10.2 points—which indicates that the intervention shifts the model's overall operating point rather than harming any particular group.

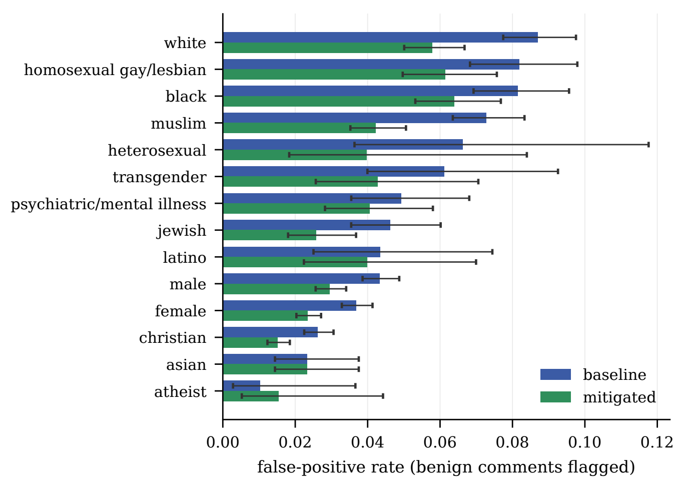

Fig. 5. False-positive rate by subgroup with Wilson 95% intervals, seed 42, for the 14 subgroups clearing the FPR support floor. Twelve improve, `asian` is unchanged, and `atheist`—the smallest and best-served group—gets worse.

M. Integrity and leakage

Table X records the checks that make the paired comparisons legitimate: within every seed, all runs share one partition and one GPU, and no model collapsed.

The split is by row rather than by text, and the corpus contains repeated comments, so between 0.98% and 1.11% of test rows occur verbatim in training. We measured the consequence instead of arguing about it: removing those rows and recomputing both arms shifts the mitigation effect by at most 0.0032 and by +0.0005 on average. Both arms share one partition, so both are inflated identically and the paired difference carries no contamination; only the absolute figures are mildly inflated, and now by a measured amount.

VII. DISCUSSION

A. The finding, stated plainly

The disparity is not a bug in one model. It is an almost linear function of the training distribution, reproducible at $r \geq 0.885$ across two backbones and eight configurations. It can be reduced—reliably, and by a measurable amount—but in this weighting family it cannot be reduced for free, and the price is paid in exactly the currency a moderation system cares about most.

The more useful finding is *why* only one of our interventions worked. Three independent lines converge on it. Ordinary class weighting makes the gap 2.6× worse. Weighting the identity-toxic cell alone makes the gap worse and safety better. Group DRO, which contains no human choice whatsoever, learns almost exactly the second of those and reproduces its failure amplified. The parity gain requires an *asymmetric* weighting of the identity-bearing non-toxic cell alone; any objective that also raises the identity-bearing toxic cell travels the same

TABLE X
PER-SEED INTEGRITY CHECKS OVER ALL 38 RUNS. A SINGLE SPLIT FINGERPRINT PER SEED IS WHAT MAKES THE CONFIGURATIONS COMPARABLE WITH EACH OTHER AND NOT ONLY EACH WITH ITS OWN BASELINE.

Seed	Runs	Split fingerprint	One partition	One GPU	Collapsed	Dup. %
42	8	`798dd7cc3c01ef5a`	yes	yes	none	1.09
43	8	`39a5c590e88707e2`	yes	yes	none	1.10
44	8	`41cd4a89437acdbd`	yes	yes	none	1.00
45	4	`435d5de1b711cd7c`	yes	yes	none	0.98
46	4	`3a7fa90ef16c138c`	yes	yes	none	1.06
47	2	`8a1a4c881597f3e9`	yes	yes	none	1.11
48	2	`c32c2cd9766f0580`	yes	yes	none	0.98
49	2	`1265af7b462697f9`	yes	yes	none	1.02

frontier backwards. A worst-group objective cannot find our solution, because the identity-toxic cell also carries high loss.

The held-out experiment (Section VI-G) qualifies how that mechanism should be described. What the weighting teaches does generalise to axes it never saw, but as a *level shift* rather than a change of ordering: the model becomes more cautious about identity-bearing language in general without re-ranking the groups against their toxic rate. The shortcut and the disparity are related but not identical—which is the conclusion the augmentation dose experiment reaches from the opposite direction, and the two agreeing is the strongest evidence we have for it.

B. What this means for a deployed system

An operator choosing among these methods faces a choice we can now state concretely. Post-processing gives the largest parity gain for no training cost but requires demographic information at inference time. Reweighting gives a slightly smaller gain, needs identity annotations only during training, and additionally makes parity far less sensitive to the threshold—a real operational advantage, since deployed thresholds move. Counterfactual augmentation, on this corpus, buys nothing at either dose. And every method that reduces the FPR gap in our study increases missed abuse; an operator who cannot accept that increase should not expect this family of methods to help.

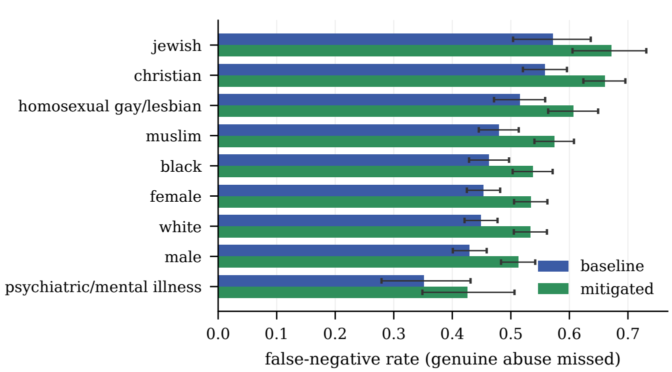

Fig. 6. False-negative rate by subgroup, seed 42, for the 9 subgroups clearing the FNR support floor. Every one is worse under the mitigation. This is the safety cost the headline table prices at 2.19.

VIII. LIMITATIONS

- **Absolute parity improved; the relative parity of the highest-support pair did not.** The `white`/`christian` ratio goes $3.32\times \rightarrow 3.83\times$. We state it that precisely because the worst/best ratio over all 14 subgroups moves the other way ($8.48\times \rightarrow 4.23\times$); a flat claim in either direction would be false.
- **One subgroup got worse.** `atheist` rose 50% in relative terms from a small base, and `asian` did not move.
- **Part of the measured FPR is annotator disagreement**, not model error—45% of a random sample, with a Wilson interval of roughly $[33\%, 58\%]$ at $n = 60$ (Section VI-K).
- **The coding has one coder**, so we report no inter-annotator agreement; a second coder on a subset would fix this cheaply, and only the baseline arm was coded.
- **Support is uneven.** 14 of 24 subgroups can carry a parity claim and only 9 a safety claim under the separate floors of Section III-B; `other_disability` has no test rows at all, so audits report 23 subgroups rather than 24.
- **Only two of the eight configurations reach eight seeds.** Four have three and two have five, which supports a direction and an interval but no significance claim. We print their sign test anyway, at its floor of 0.25, so that $3/3$ is never mistaken for evidence.
- **The worst-affected subgroup changes** between arms (`white` $\rightarrow$ `black`), and we do not explain why.
- **Attribution is not causation.** Integrated Gradients [19] measures local input sensitivity along a path from a baseline input; we report the identity-token attribution share ($0.1164 \rightarrow 0.1001$, -14% over 20 paired explanations) as supporting analysis only, never as evidence about how the model reasons.
- **Scope.** One corpus, one language, one architecture family, and binary coarse identity categories that real identity does not respect.

IX. CONCLUSION AND FUTURE WORK

We compared bias mitigations at all three points of the pipeline on one corpus under one recipe, with replication and pre-registered outcomes. Identity-aware loss reweighting reduces the subgroup FPR gap by 0.0520 over eight seeds ($p = 0.0078$) and flattens the shortcut that produces the gap by

44.6%, at a cost of 2.19 points of missed abuse per point of gap closed. Per-subgroup thresholds do better on parity and worse on deployability. Counterfactual augmentation does nothing at either dose—while measurably removing the lexical shortcut, which shows the shortcut and the disparity are not the same thing. The learned-weight version of our own objective moves the gap backwards and, in doing so, identifies the asymmetry that makes the working method work.

Three directions follow. A second coder on the error sample would put an agreement statistic on the 45% figure. Extending the held-out experiment to more axis splits would turn a suggestive transfer result into a tested one. And the frontier itself invites a direct attack: the coupling we document is a property of a weighting family, and whether an intervention outside that family can close the gap without paying in safety is, on this evidence, still open.

Tools and Acknowledgements

AI assistance (large-language-model coding assistants) was used during this project for: refactoring and documenting source code; drafting and copy-editing the English prose of this report; searching and summarising related literature; and generating boilerplate for plotting and table export. All experimental design decisions, the pre-registration of the outcome thresholds, the choice of metrics, controls and support floors, the manual error coding, and the interpretation of every result reported here are the authors' own. No AI system was used to generate, alter or select any experimental result. Every line of the released code has been read by the authors, and each author is able to explain and defend any part of the system.

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References

[1] L. Dixon, J. Li, J. Sorensen, N. Thain, and L. Vasserman, "Measuring and mitigating unintended bias in text classification," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society (AIES)*, New Orleans, LA, USA, 2018, pp. 67–73.

[2] D. Borkan, L. Dixon, J. Sorensen, N. Thain, and L. Vasserman, "Nuanced metrics for measuring unintended bias with real data for text classification," in *Companion Proc. World Wide Web Conf. (WWW)*, San Francisco, CA, USA, 2019, pp. 491–500.

[3] M. Sap, D. Card, S. Gabriel, Y. Choi, and N. A. Smith, "The risk of racial bias in hate speech detection," in *Proc. 57th Annu. Meeting Assoc. for Computational Linguistics (ACL)*, Florence, Italy, 2019, pp. 1668–1678.

[4] J. H. Park, J. Shin, and P. Fung, "Reducing gender bias in abusive language detection," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Brussels, Belgium, 2018, pp. 2799–2804.

[5] T. Garg, S. Masud, T. Suresh, and T. Chakraborty, "Handling bias in toxic speech detection: A survey," *ACM Comput. Surv.*, vol. 55, no. 13s, art. 264, pp. 1–32, 2023.

[6] S. Gupta, V. Kovatchev, A. Das, M. De-Arteaga, and M. Lease, "Finding Pareto trade-offs in fair and accurate detection of toxic speech," *Information Research*, vol. 30, iConf., pp. 123–141, 2025.

[7] S. Garg, V. Perot, N. Limtiaco, A. Taly, E. H. Chi, and A. Beutel, "Counterfactual fairness in text classification through robustness," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society (AIES)*, Honolulu, HI, USA, 2019, pp. 219–226.

[8] J. Lu, B. Xu, X. Zhang, K. Liu, D. Zhang, L. Yang, and H. Lin, "Take its essence, discard its dross! Debiasing for toxic language detection via counterfactual causal effect," 2024, arXiv:2406.00983.

[9] B. H. Zhang, B. Lemoine, and M. Mitchell, "Mitigating unwanted biases with adversarial learning," in *Proc. AAAI/ACM Conf. AI, Ethics, and Society (AIES)*, New Orleans, LA, USA, 2018, pp. 335–340.

[10] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.

[11] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems (NIPS)*, Barcelona, Spain, 2016, pp. 3315–3323.

[12] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," 2019, arXiv:1910.01108.

[13] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," 2019, arXiv:1907.11692.

[14] cjadams, D. Borkan, inversion, J. Sorensen, L. Dixon, L. Vasserman, and N. Thain, "Jigsaw unintended bias in toxicity classification," Kaggle competition, 2019. Dataset licensed CC0 1.0. [Online]. Available: https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification

[15] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. Conf. Empirical Methods in Natural Language Processing: System Demonstrations*, 2020, pp. 38–45.

[16] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 8024–8035.

[17] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[18] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[19] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. Int. Conf. Machine Learning (ICML)*, Sydney, Australia, 2017, pp. 3319–3328.